HOW TO CRACK THE INTERVIEW

ST Logistics — Senior Manager

Data, Analytics & AI

Strategy & Transformation

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A Brutally Practical Guide

WHAT Role Decoding **WHY** Business Impact **HOW** Technical Prep **FLOW** Interview Rounds + 90-Day Plan

WHAT THEY WANT

The "T-Shape" Profile

BROAD — Horizontal Skills

- Full-stack engineering (APIs, frontend)
- Cloud architecture (Azure, Databricks, Fabric)
- CI/CD, containerization, version control
- Stakeholder communication & storytelling
- Leadership & team management
- Data governance & PDPA compliance

DEEP — Vertical Expertise

- Computer vision models (YOLO, ResNet)
- Forecasting (Prophet, XGBoost, LSTM)
- Route optimisation algorithms
- Real-time + batch data pipelines
- Model deployment & scaling (MLOps)
- Data Analytics & AI/ML at scale

Not a pure data scientist. Not a pure engineer. A unicorn who does both + leads a team + presents to C-suite.

WHICH TECH TO FOCUS FIRST

The Priority Stack

TIER 1 — 60% of Prep Time

"Asked in every round"

1. **SQL (Advanced)**
   Window functions, CTEs, query optimization — they WILL give you a live SQL test
2. **Python + Pandas**
3. **PySpark / Spark**
4. **Azure (ADF, ADLS, Synapse, Azure ML)**
5. **Databricks + Fabric (Unity Catalog, Delta Lake)**

TIER 2 — 25% of Prep Time

"Deep-dive in Round 2-3"

6. **ML/AI Frameworks (PyTorch, CV, Time Series)**
7. **MLOps (Docker, CI/CD, MLflow)**
8. **API Dev (FastAPI)**
9. **Data Governance (PDPA!)**

TIER 3 — 15% of Prep Time

"Mention to impress"

10. **LLMs (Hugging Face, OpenAI, RAG)**
11. **Route Optimisation (OR-Tools)**
12. **Full-stack (REST)**
13. **Streaming (Kafka)**

Memory Trick: SQL → Python → Spark → Cloud → ML → Deploy — the data lifecycle itself

THE INTERVIEW FLOW

4 Rounds, 4 Skill Domains

R1	**HR + Hiring Manager** Culture Fit, Leadership, Motivation	45 min
R2	**Technical Lead / Data Team** Live SQL, Python, Architecture	90 min
R3	**VP / Director Level** System Design, Business Value, ROI	60 min
R4	**C-Suite (CTO/COO)** Vision, Transformation, ROI	45 min

ROUND 1

"Why ST Logistics? It's not a tech company."

1. FLIP THE NARRATIVE

"Not a tech company" = feature, not a bug. Constraint-rich + data-rich = every model has real-world impact.

2. SHOW DOMAIN VALUE

2% route optimization = $100K+ saved/yr
15% forecast improvement = $300K inventory reduction
CV damage detection = 60% faster claims

3. CAREER AMBITION

Building data culture from scratch = leadership story. Proving AI ROI in operations = C-suite credibility.

ROUND 1

Convince the C-suite — STAR Framework

S

SITUATION

Ops team used spreadsheets for demand forecasting. ~60% accuracy.

T

TASK

Secure $150K budget for Azure + Databricks + 1 hire.

A

ACTION

3-Meeting Strategy:
Meeting 1: Show pain ($2.3M excess inventory)
Meeting 2: Show proof (2-week POC: 82% vs 60%)
Meeting 3: Show roadmap (phased 3-6-12 months)

R

RESULT

Budget secured. Deployed in 4 months. Year 1 savings: $920K

KEY: Quantify everything. Use dollar amounts. Show phased approach. Never burn budget all at once.

ROUND 1

Managing Morale When Under-Resourced

1

PROTECT THE TEAM'S TIME

I am the shield between C-suite and my engineers. Every ad-hoc request goes through me. I negotiate scope and timeline before it hits the team.

2

MAKE WINS VISIBLE

Weekly "Demo Friday" — 15 minutes, someone shows what they built. I invite stakeholders. My team gets the credit, not me. Recognition beats pizza parties.

3

INVEST IN GROWTH

Each team member gets 4 hrs/week for learning. Certification prep, side projects. Not optional — I protect that time like production uptime.

ROUND 2 — TECHNICAL

Live SQL: Top 3 Warehouses by Delay

```sql
WITH delivery_delays AS (
  SELECT warehouse_id,
    DATEDIFF(day,scheduled,actual)
    - 2*DATEDIFF(week,scheduled,actual)
    AS business_day_delay
  FROM deliveries
  WHERE actual >= DATEADD(day,-90,GETDATE())
    AND status = 'COMPLETED'
),
warehouse_stats AS (
  SELECT warehouse_id,
    AVG(business_day_delay) AS avg_delay,
    RANK() OVER (ORDER BY
      AVG(business_day_delay) DESC) AS rnk
  FROM delivery_delays GROUP BY warehouse_id
)
SELECT * FROM warehouse_stats WHERE rnk <= 3;
```

Narration Tips

Use CTEs
Self-documenting code

Business Days
Exclude weekends (logistics!)

Add P95
Averages hide outliers

RANK() not ROW_NUMBER()
Ties both appear in top 3

ROUND 2 — TECHNICAL

Real-Time Shipment Tracking Pipeline

DATA SOURCES

IoT/GPS · Partner APIs · Barcode Scans

▼

INGESTION

Azure Event Hubs (Kafka-compatible)
Partitioned by region · 1M events/min

▼

HOT PATH (Real-time)

Spark Structured Streaming
30-second micro-batch windows
Watermark: 10 min (late events)

COLD PATH (Batch)

ADF → ADLS Gen2 → Delta Lake
Bronze / Silver / Gold layers
Daily aggregates for reporting

▼

DASHBOARD

Power BI / Fabric (DirectLake mode) — Auto-refresh <5 min — Live shipment map · Delay alerts · Warehouse KPIs

Delta Lake for ACID transactions (no duplicates in logistics) · DirectLake = no import scheduling needed

ROUND 2 — CORE CONCEPT

Medallion Architecture — Memorize This

BRONZE

Raw, Append-only

Data as-is from source
Schema-on-read
Full audit trail

SILVER

Cleaned, Typed

Deduplicated & validated
Strong types applied
Business logic joins

GOLD

Business-ready

Star schema aggregates
KPI tables for Power BI
ML-ready feature tables

Storage

ADLS Gen2 + Delta Lake

Governance

Unity Catalog + Purview

Consumed By

Power BI (DirectLake), FastAPI

Why This Matters

ACID transactions prevent duplicates in logistics data. Schema enforcement catches bad data before it reaches Gold. Time travel enables full auditability.

Computer Vision + Demand Forecasting

Package Damage Detection (CV)

Phase 1: Data
5K+ labeled images · 5 classes
Phase 2: Model
YOLOv8 / ResNet50 transfer learning
Target: mAP > 0.85
Phase 3: Deploy
Edge (Jetson) + Cloud sync
<100ms inference on conveyor belt
Phase 4: Business
Auto-flag WMS → Carrier liability report

Demand Forecasting (Ensemble)

Features
Lags (t-1, t-7, t-30) · Rolling stats
Calendar (CNY, monsoon, holidays)
Weather · Product category · Region
Model Ensemble
Prophet (0.3) + XGBoost (0.5) + LSTM (0.2)
Target: MAPE < 15%
Deployment
Databricks scheduled daily reforecast
FastAPI: /predict/demand?warehouse=SG-01

ROUND 2 — GOVERNANCE

PII Incident Response + PDPA Compliance

HOUR 0	**CONTAIN** Stop pipeline · Identify blast radius · Revoke access
HOUR 1-4	**NOTIFY** DPO + IT Security · PDPA: notify PDPC within 3 days if significant
DAY 1-3	**REMEDIATE** Hash PII (SHA-256 + salt) · Purge Gold tables · Add PII scanner
WEEK 1-2	**PREVENT** Column-level access (Unity Catalog) · CI/CD quality gates · Training

PDPA Singapore: Purpose limitation · DPO required · NRIC + phone = PII · 3-day breach notification

ROUND 3 — BUSINESS VALUE

"Show Me the ROI" — $500K Investment

Use Case	Annual Value	How Measured
Demand Forecasting	**$320K**	40% less excess inventory
Route Optimization	**$180K**	12% less fuel + overtime
Damage Detection (CV)	**$95K**	60% faster claims
Automated Reporting	**$75K**	200 analyst-hrs/mo freed
Real-time Dashboard	**$60K**	25% fewer escalation calls
TOTAL ANNUAL VALUE	**$730K**	**Year 1 ROI: 46% | Year 2+: >100%**

"We built the platform. Every additional use case now costs 70% less to deploy. Year two ROI will be north of 100%."

ROUND 4 — VISION

LLM Integration in Logistics

Document Processing

Bills of Lading, Customs forms, PODs
OCR + GPT-4 structured extraction
Auto-populate WMS fields, flag for review

Natural Language Queries

50 people need data, only 3 can write SQL
Text-to-SQL agent over Gold tables
"On-time delivery rate for electronics?" → 92.3%

Anomaly Explanation

Dashboard spike → LLM pulls correlated data
Generates narrative: "Delay spike correlates with Carrier X vehicle shortage + client Y flash sale"

Tech: Azure OpenAI + Azure Doc Intelligence for OCR · Hugging Face for custom fine-tuning · RAG for grounded answers

ROUND 4 — CURVEBALL

Day 1 at ST Logistics — 90-Day Plan

DAYS 1-30 **LISTEN & AUDIT**	1-on-1 with every team member & stakeholder Audit pipelines: what runs, breaks, is manual? Sample 10 tables → data quality scorecard "State of Data" report → identify 3 quick wins
DAYS 31-60 **QUICK WINS + FOUNDATION**	Execute 3 quick wins (fix pipeline, automate report) Set up CI/CD for data pipelines Basic data quality monitoring Weekly "Data Office Hours" for stakeholders
DAYS 61-90 **FIRST BIG PROJECT**	Launch demand forecasting (highest ROI) Data governance v1 (ownership, catalog, PII rules) Present roadmap to C-suite with ROI projections Set quarterly OKRs for the data team

HOW TO REMEMBER EVERYTHING

Memory Palace + Daily Drill

FRONT DOOR

R1 Culture

ST Logistics doorbell → warehouse inside
3 leadership stories pinned to wall

KITCHEN

R2 Technical

Counter = SQL queries on cutting board
Stove = Spark cooking Bronze→Silver→Gold
Fridge = ADLS Gen2 cold storage

LIVING ROOM

R3 Design

TV = architecture diagram playing
Bookshelf = layers (Ingest, Store, Compute)
Coffee table = ROI calculations

BEDROOM

R4 Vision

Pillow = 90-day plan tucked underneath
Alarm clock = "Day 1 — what do you do?"

5-4-3-2-1 Daily Drill (2 weeks pre-interview)

5 = Draw Medallion arch | 4 = Recite 4 AI use cases | 3 = Write 3 SQL patterns | 2 = Explain 2 models E2E | 1 = Tell 1 story with $$$

CHEAT SHEETS

Acronyms to Carry Into the Room

DELTA

Data Platform

D
Databricks (compute)
E
Event Hubs (streaming)
L
Lake (ADLS Gen2 + Delta)
T
Transformation (Spark)
A
APIs (FastAPI)

CRISP

Answer Any Tech Q

C
Context ("In logistics...")
R
Requirements (constraints)
I
Implementation (build it)
S
Scale (handles X because...)
P
Production (deploy how)

PDPA

Singapore Data Law

P
Purpose limitation
D
DPO required
P
Personal data (NRIC=PII)
A
Accountability (prove it)

Breach: 3-day notification

Key Flashcards

Medallion arch?	Bronze→Silver→Gold on Delta Lake	**Prophet vs XGBoost?**	Seasonality vs non-linear. Ensemble both.
Unity Catalog?	Databricks governance: ownership + lineage	**DirectLake?**	Power BI reads Delta directly, no import

FINAL PREP

Night Before Checklist

- ☐ Laptop charged, demo code ready to screen-share
- ☐ Architecture diagrams printed (or iPad ready)
- ☐ 3 leadership stories rehearsed WITH NUMBERS
- ☐ 90-day plan on one page
- ☐ ST Logistics annual report skimmed (clients, revenue)
- ☐ 5 questions for them written down
- ☐ Good night's sleep

"They want a unicorn. Unicorns are just people who prepared more than everyone else."